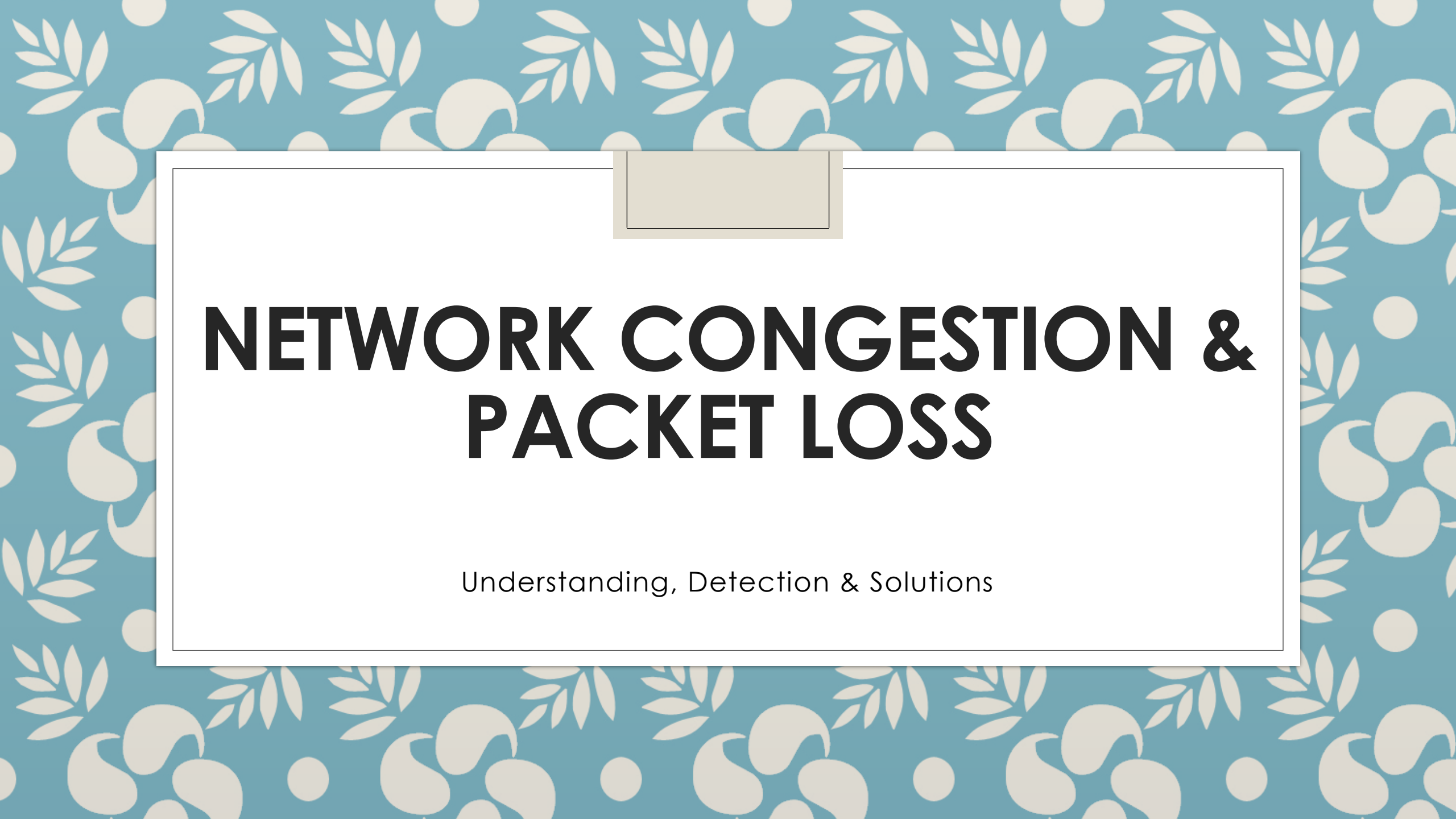




NETWORK CONGESTION & PACKET LOSS

Understanding, Detection & Solutions

WHAT IS OUR AGENDA?

- Introduction to Computer Networking Basics
- Explanation of Network Congestion
- Understanding Packet Loss
- Detection and Monitoring Techniques
- Solutions and Prevention Methods

What is Computer Networking?

- Computer networking connects multiple devices to share data and resources.
- It enables communication between phones, computers, servers and cloud systems.
- Data travels through networks in the form of digital signals and packets.
- Modern networks handle large amounts of traffic every second.
- Network speed and reliability directly affect user experience.

Understanding Network Congestion

- Network congestion is like a traffic jam.
- It occurs when data traffic exceeds network capacity and network is not able to handle the requests.
- Routers and switches receive more packets than they can process.
- Incoming packets are stored temporarily in queues.
- When queues become full, packets are delayed or dropped.
- This results in slow speed, high latency, and unstable connections.

WHAT PROBLEMS DO WE FACE ?

- Data - transferred in form of packets
- Packets are delayed, queued, or dropped due to limited resources.
- Leads to reduced speed, video buffering, and interrupted connections.
- Affects both individual users and organizational operations.

Main Causes of Congestion (Part 1)

- **High Traffic Volume:** Too many users or devices sending data simultaneously
- **Limited Bandwidth:** Not enough capacity to handle all requests
- **Bandwidth Hogs:** Some devices consuming excessive network resources (4k video streaming, IPL streamings etc..)
- **Poor Network Design:** Inefficient network structure and planning

Main Causes of Congestion (Part 2)

- **Outdated Hardware:** Old routers and switches cannot handle modern speeds
- **Faulty Equipment:** Damaged cables or malfunctioning network devices
- **Broadcast Storms:** Sudden spike in network requests
- **Poor Configuration:** Incorrect network settings and management

What is Packet Loss?

- Data is broken into packets.
- Each packet travels through the network independently.
- Packet loss is when data packets fail to reach their destination during transmission.
- Lost packets must be retransmitted, increasing delay.
- Even a small amount of packet loss can significantly reduce network performance.

Why Does Packet Loss Happen?

- **Network Congestion:** Routers drop packets when queues are full
- **Hardware Failures:** Faulty network devices losing data
- **Cable Issues:** Damaged or loose network cables
- **Software Bugs:** Errors in routing protocols or firmware

Impact on Network Performance

- **For Users**

- Slow loading of websites and applications
- Frequent buffering during video streaming
- Interrupted voice and video calls
- Lag and disconnections in online gaming

- **For Businesses**

- Service downtime and system unavailability
- Reduced employee productivity
- Poor customer experience
- Financial losses and reduced revenue

Technical Performance Issues

- **High Latency:** Increased delay in packet delivery time
- **Reduced Throughput:** Lower data transfer rates
- **Jitter:** Inconsistent packet arrival times
- **Connection Timeouts:** Failed connections due to delays

How to Detect Congestion & Packet Loss

- **Network Monitoring Tools:** Real-time traffic analysis software
- **Packet Sniffers:** Tools like Wireshark to capture packets
- **Performance Metrics:** Track latency, throughput, packet loss rate
- **Traffic Analysis:** Identify bandwidth usage patterns

Prevention Strategies (Part 1)

- **Increase Bandwidth:** Upgrade to higher capacity connections
- **Load Balancing:** Distribute traffic across multiple paths
- **Traffic Shaping:** Control and prioritize data flow
- **Network Segmentation:** Divide network into smaller subnets
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Prevention Strategies (Part 2)

- **Quality of Service (QoS):** Prioritize critical applications
- **Efficient Routing:** Optimize packet paths using better protocols
- **Payload Compression:** Reduce data size before transmission
- **Hardware Upgrades:** Replace outdated network equipment

Quality of Service (QoS)

- QoS is a technique to manage network resources and prioritize important traffic.
- Video calls and VoIP get higher priority than downloads
- Can reduce packet loss by up to 25%
- Ensures critical business applications work smoothly

Network Optimization Best Practices

- Regular network monitoring and maintenance
- Update firmware and software regularly
- Implement proper network security measures
- Plan for future capacity needs

Real-World Solutions

- **SD-WAN = Software-Defined Wide Area Network**
- It is a technology where **software, not hardware**, decides how data should travel across the network.
- Instead of fixed routes, the system **dynamically chooses the best path.**
- **TCP = Transmission Control Protocol**
- TCP controls:
 - How data is sent
 - How errors are corrected
 - How speed is adjusted
- It is the “traffic controller” of data transfer.

Key Takeaways

- Network congestion is a common challenge in modern networks
- Proper planning and monitoring can prevent most issues
- Multiple solutions exist - choose based on your needs
- Regular maintenance ensures long-term network health

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THANKYOU !